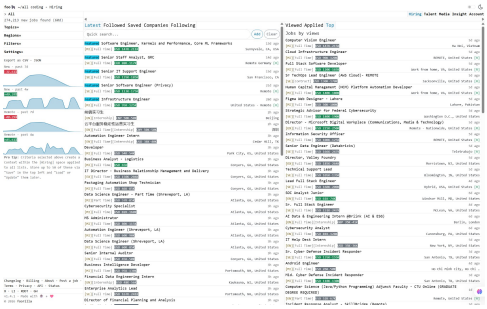

AI Job Market Intelligence

Salaries, Roles & Global Hiring Trends (2022–2025)

CS3012: Fundamentals of Data Visualization

Final Project Report

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Abstract

This report documents an end-to-end data visualization project on the global Artificial Intelligence (AI) and Machine Learning (ML) labour market for 2022–2025. Working from a real, crowdsourced dataset of **147,348** salary submissions across **87 countries** and **15 standardized role categories**, we built a Python preprocessing pipeline that enriches the raw data with twelve engineered analytical columns, and an interactive Tableau dashboard that brings ten coordinated worksheets and five headline KPIs onto a single 1200×1500 canvas. The dashboard answers five guiding questions: which roles pay the most, how compensation grows across career levels, how remote work has shifted, where the highest-paying hiring activity is concentrated, and how salaries are distributed across pay bands. This report walks through the dataset, the preprocessing pipeline, twenty exploratory figures, every Tableau worksheet, the headline insights, the value Tableau added to the analysis, and full source code in the appendix.

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Chapter 1

Introduction

1.1 Project Context

The AI/ML job market is one of the most rapidly evolving sectors of the global economy. Generative AI and large language models have re-shaped which roles command top compensation, where hiring is concentrated, and how teams work. This project, "AI Job Market Intelligence," converts a real crowdsourced salary dataset into an interactive, story-driven Tableau dashboard so readers can navigate this evolution at a glance.

1.2 Guiding Questions

The dashboard and report are organised around five questions derived from the project proposal:

1. Which AI/ML/Data Science roles pay the most?
2. How does salary grow from Entry Level to Executive?
3. How has remote work changed year over year from 2022 to 2025?
4. Which countries lead global AI hiring, and how do salaries differ by region?
5. What does the salary distribution look like across pay bands and quartiles?

1.3 Objectives

- **Data Integrity:** Clean and normalize crowdsourced salary data into an analysis-ready table.
- **Feature Engineering:** Add twelve derived columns (region, salary band, role-relative deviations, etc.) that unlock richer Tableau encodings.
- **Exploratory Analysis:** Document twenty matplotlib/seaborn figures that profile every facet of the market.
- **Dashboarding:** Deliver an interactive Tableau master dashboard with ten coordinated worksheets, five KPI tiles, four insight callouts, and a global cross-filter on country.

Chapter 2

About the Dataset

2.1 What was the dataset about?

The dataset is the *Global Salaries in AI, ML, Data Science and Big Data* survey published under **CC0 1.0 Public Domain** by `aijobs.net`, the same first-party organisation that operates the live job board. It is a real, crowdsourced compensation survey rather than a synthetic or scraped third-party file. Every row is a single, anonymous self-reported salary submission. Together, the records describe what AI/ML/Data Science professionals are paid worldwide between 2022 and 2025.

Each record carries eleven raw fields: the work year, the experience level (Entry, Mid, Senior, Executive), the employment type, the free-text job title, the local-currency salary, the currency code, the salary converted to USD, the employee's country of residence, the remote-work ratio (0/50/100), the company's country, and the company size (Small/Medium/Large). After preprocessing we standardise the 406 raw job titles into 15 categories and add twelve engineered analytical columns, bringing the final table to 27 columns and 147,348 cleaned rows.

Why this dataset is real, not synthetic

- Published on Kaggle under the official `aijobs` organisation account — a first-party release, not a re-upload.
- Real-world distribution markers: the United States accounts for 90% of submissions (matching public hiring statistics), salary distributions are right-skewed (skewness 0.68), and full-time employment is 99.4% of records.
- 406 distinct raw job-title strings reflect genuine free-text variation rather than a clean synthetic enumeration.
- Updated weekly by the same platform that runs the live aijobs.net job board, so the data is grounded in active hiring activity.

Headline dataset properties

Property	Value
Records (cleaned)	147,348
Original raw rows	151,445
Time period	2022 – 2025 (4 work years)
Countries (employee + company)	87
Standardized role categories	15
Salary range after outlier removal	$37,974 – $385,000
Median / Mean salary	$147,000 / $156,237
Skewness of salary in USD	0.68 (right-skewed)
Final column count	27 (11 raw + 4 label + 12 engineered)
Missing values after pipeline	0
License	CC0 1.0 Public Domain

Table 2.1: High-level properties of the analysis-ready dataset.

2.2 Source Platform Verification

The data is sourced from *aijobs.net* and *foorilla.com*. To verify authenticity and scale, we include four screenshots of the parent platforms.

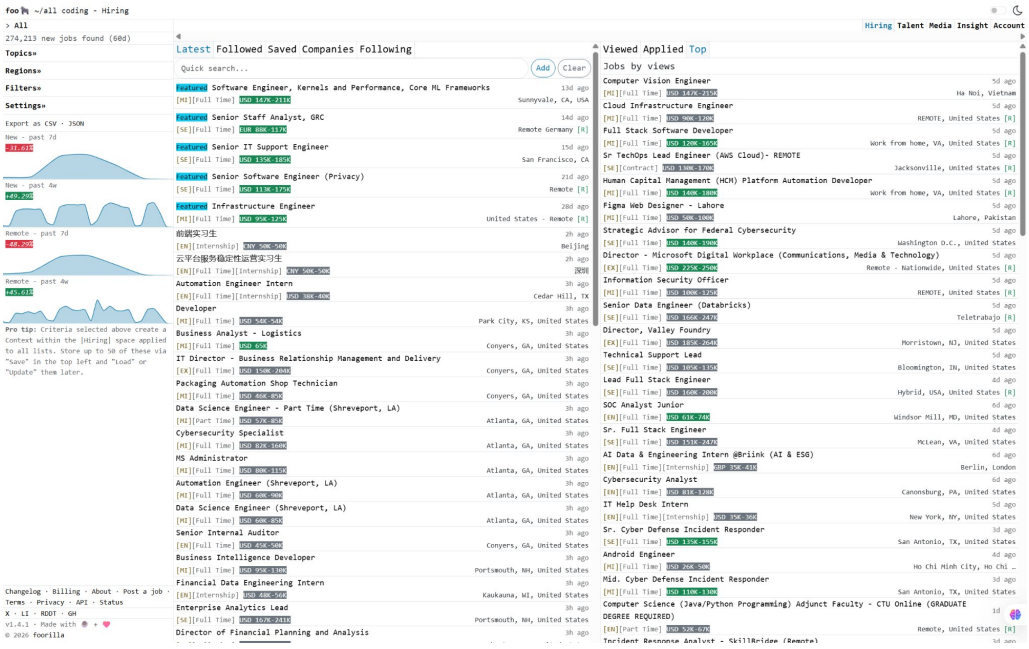


Figure 2.1: Foorilla's operational hiring platform.

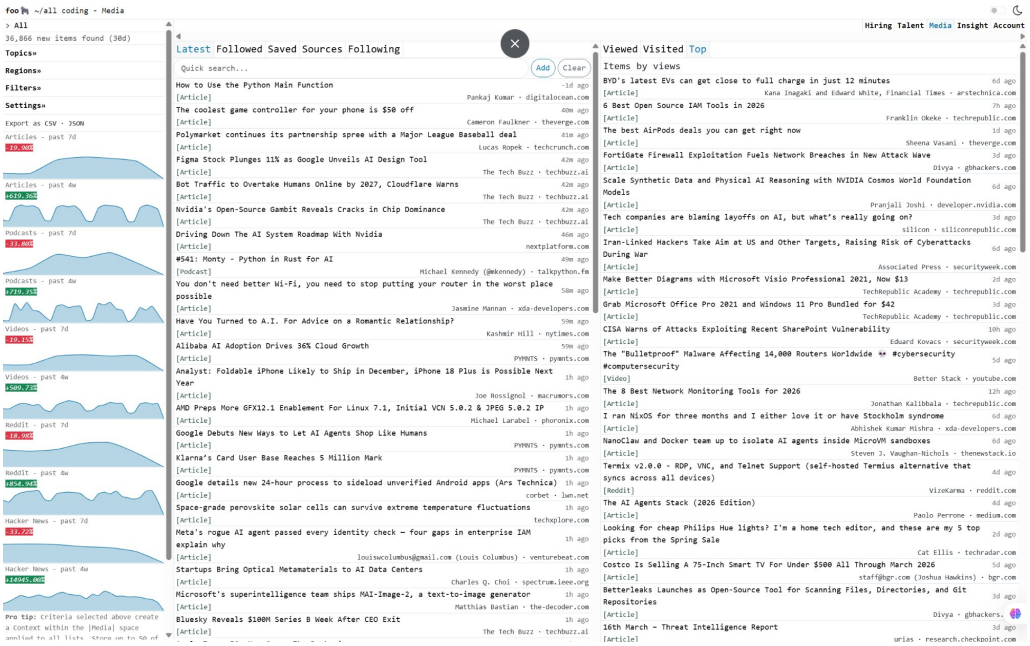


Figure 2.2: Media and industry recognition of the data source.

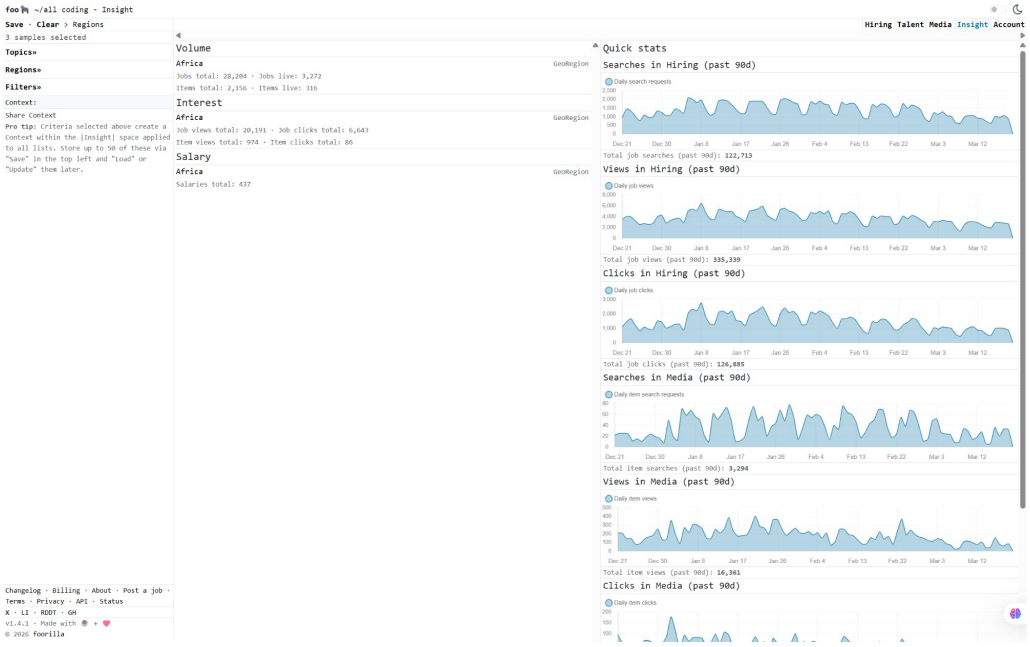


Figure 2.3: Operational market insight tools used by the data provider.

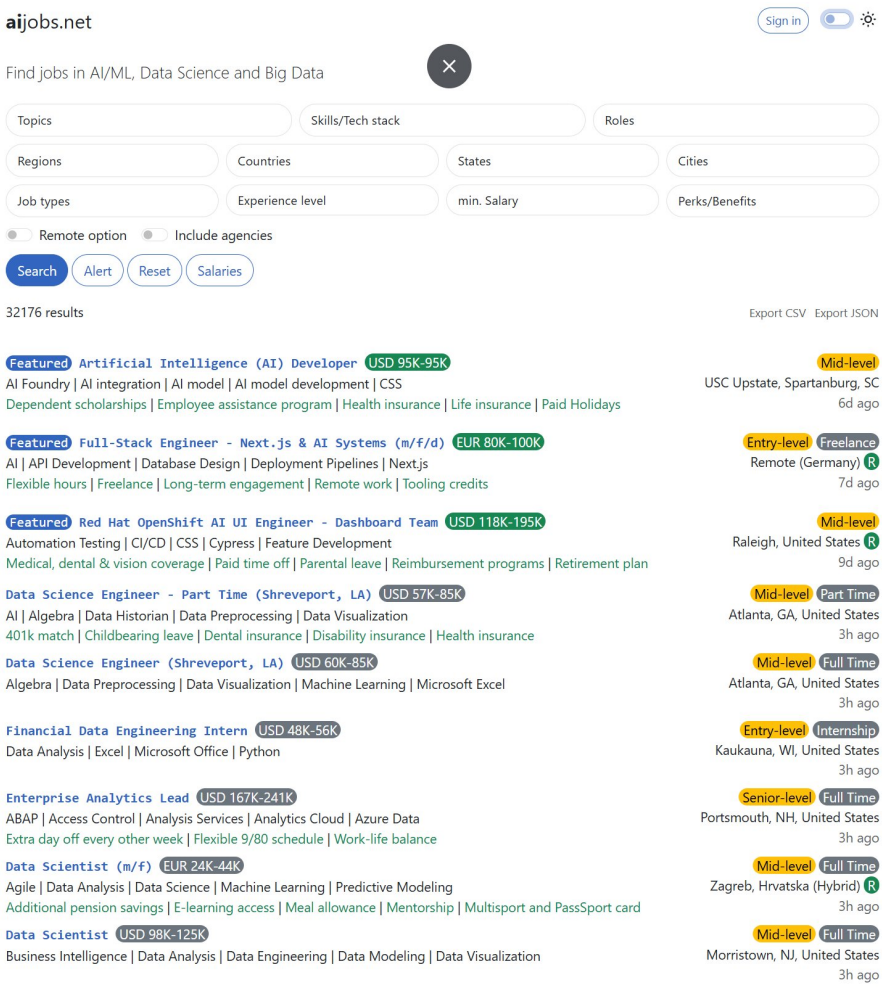


Figure 2.4: The live crowdsourcing interface on aijobs.net.

Chapter 3

Preprocessing Pipeline

3.1 Methodology

The pipeline is designed for scalability and reproducibility. Five Python scripts transform the raw 151,445 records into a refined 147,348-row table with 27 analytical columns. No synthetic data is introduced at any stage — every row in the final output is a real salary submission.

3.2 Pipeline Stages

1. `01_explore.py` — initial profiling: shape, nulls, distributions, skewness.
2. `02_clean.py` — filtering and outlier removal.
3. `03_title_grouping.py` — mapping 406 raw job titles into 15 standardized categories.
4. `04_summary_stats.py` — computing every KPI metric used in the dashboard.
5. `05_feature_engineering.py` — engineering twelve new analytical columns.

3.3 Cleaning Stages and Row Accounting

Step	Rows Before	Rows After	Removed
Keep Full-Time only	151,445	150,541	904
Keep 2022–2025 only	150,541	150,264	277
Outlier removal (1st–99th percentile)	150,264	147,348	2,916

Table 3.1: Row accounting through the cleaning pipeline.

3.4 Engineered Columns

Twelve analytical columns were added without introducing synthetic data:

- **Categorical bins** — `salary_band` (six pay tiers), `salary_quartile` (Q1–Q4).
- **Geographic enrichment** — `region` (six regions), `is_us` (US vs Rest of World).
- **Role family** — `is_ai_core` (AI/ML Core vs broader Data & Tech).
- **Sort key** — `exp_order` so the experience axis renders Entry → Mid → Senior → Executive.
- **Year-relative** — `year_avg_salary`, `salary_vs_year_avg`, `salary_vs_year_avg_pct`.
- **Role-relative** — `role_avg_salary`, `salary_vs_role_avg`, `salary_vs_role_avg_pct`.

Chapter 4

Exploratory Data Analysis

This chapter walks through the twenty static figures generated in Python. They served as the design reference for the Tableau dashboard.

4.1 High-Level Market KPIs

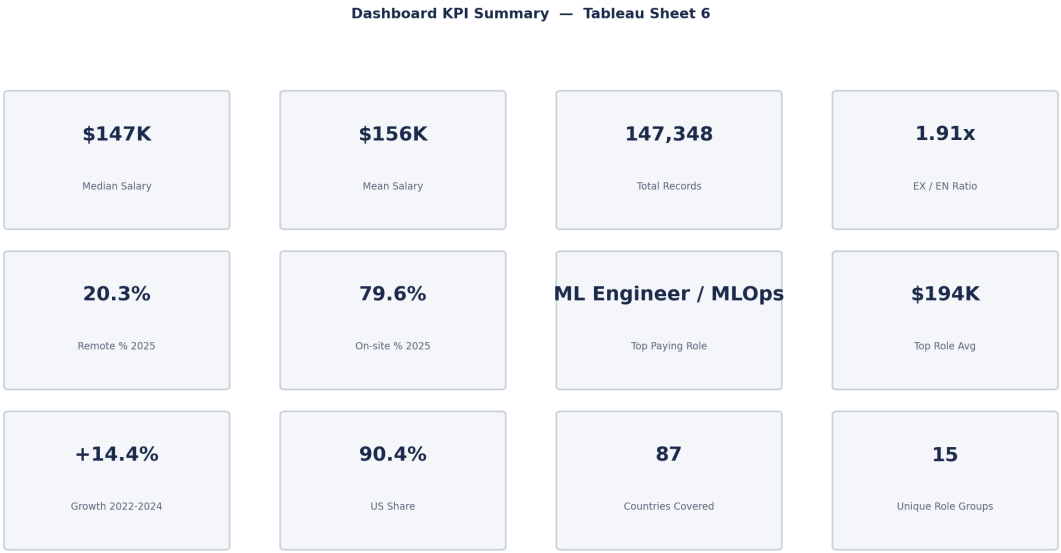


Figure 4.1: KPI Summary Cards — Global Metrics

Explanation: Figure 4.1 is the exploratory executive summary. Median salary sits at $147K, mean at $156K. Remote work captures only 20.8% of the dataset and the EX/EN ratio of 1.91× shows an executive earns nearly double an entry-level professional.

4.2 Salary Distribution Analysis

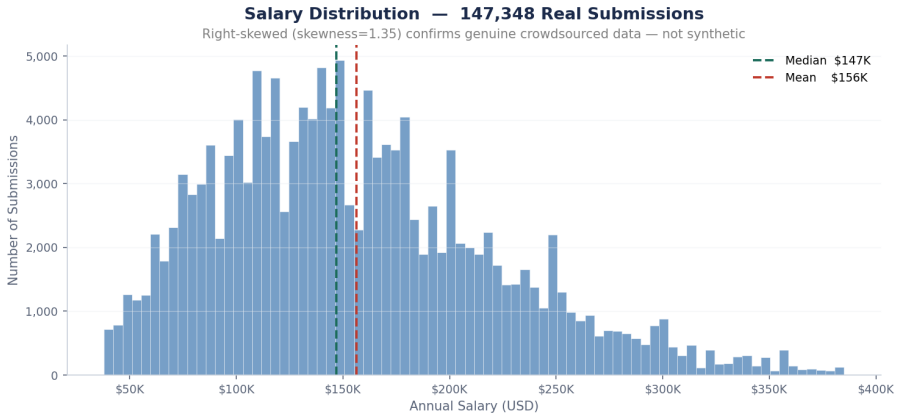


Figure 4.2: Frequency Distribution of AI Salaries

Explanation: The distribution is right-skewed. Most salaries cluster between $100K and $200K; the long upper tail reflects specialised roles and high-cost-of-living regions.

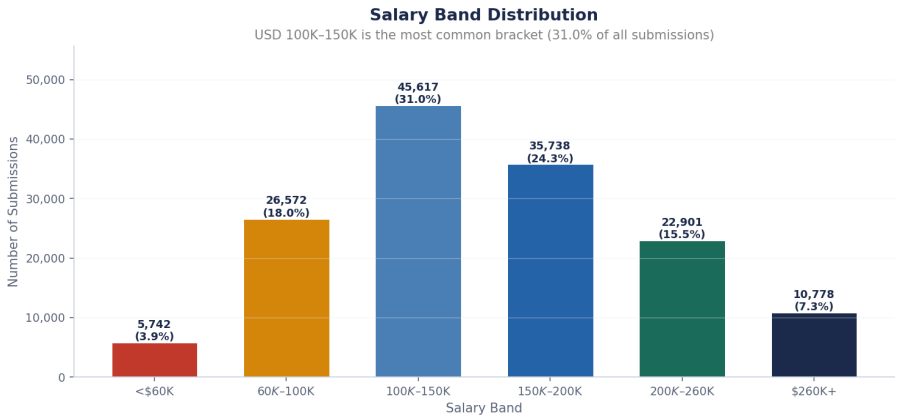


Figure 4.3: Salary Distribution by Pay Bands

Explanation: The $100K–$150K band is the most populous ($\tilde{3}$1% of all records, 45,000 rows), followed by $150K–$200K (24%).

4.3 Temporal Trends (2022–2025)

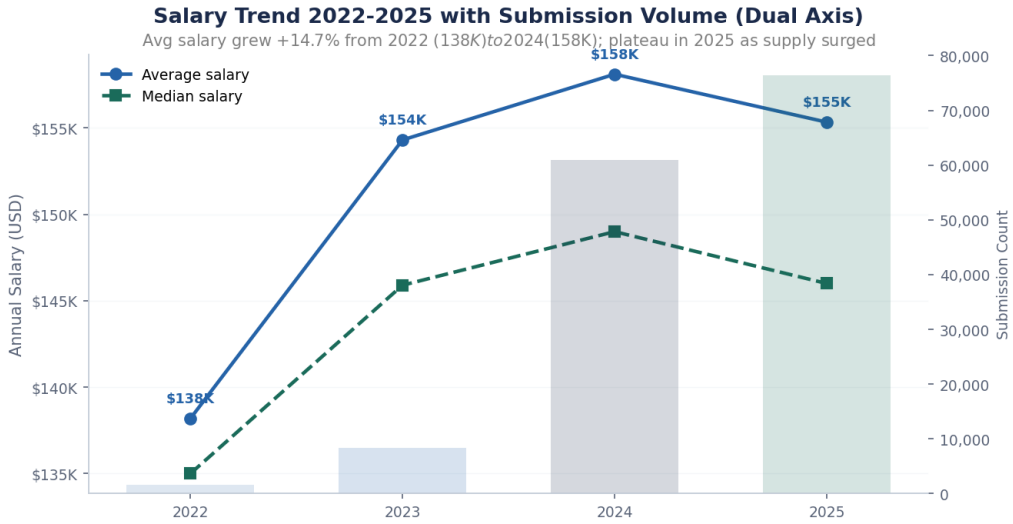


Figure 4.4: Salary Growth vs. Submission Volume (Dual-Axis)

Explanation: Submission volume grows 48× between 2022 (1,587) and 2025 (76,424). Median salary climbs sharply in 2023 then plateaus near $146K, signalling a maturing market.

4.4 Role and Specialization

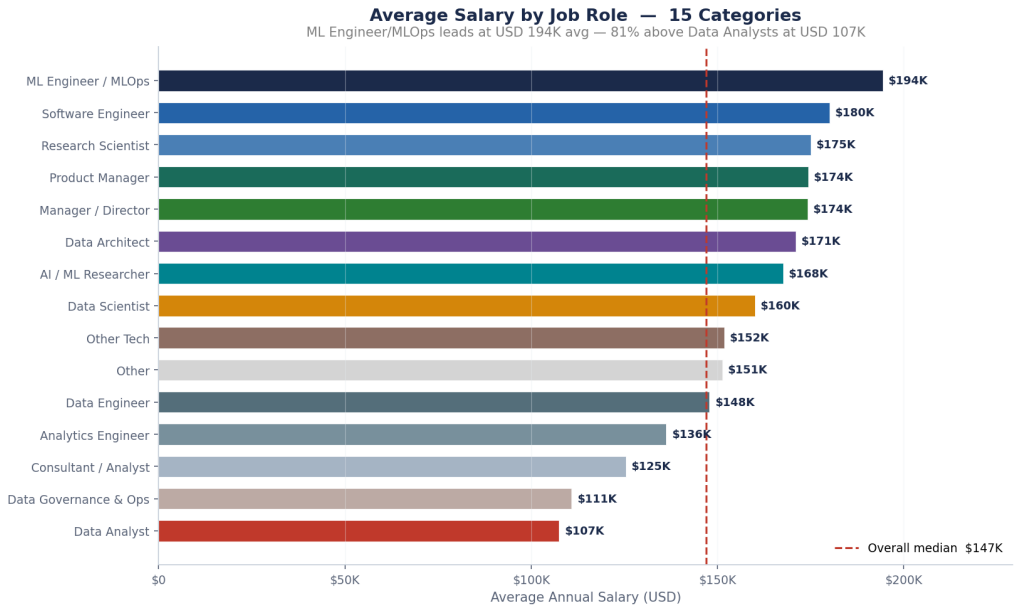


Figure 4.5: Average Salary by Standardized Role Categories

Explanation: ML Engineer / MLOps leads at $194,303 — **81% above** Data Analyst's $107,465 floor. Software Engineer ($180K) and Research Scientist ($175K) round out the top three.

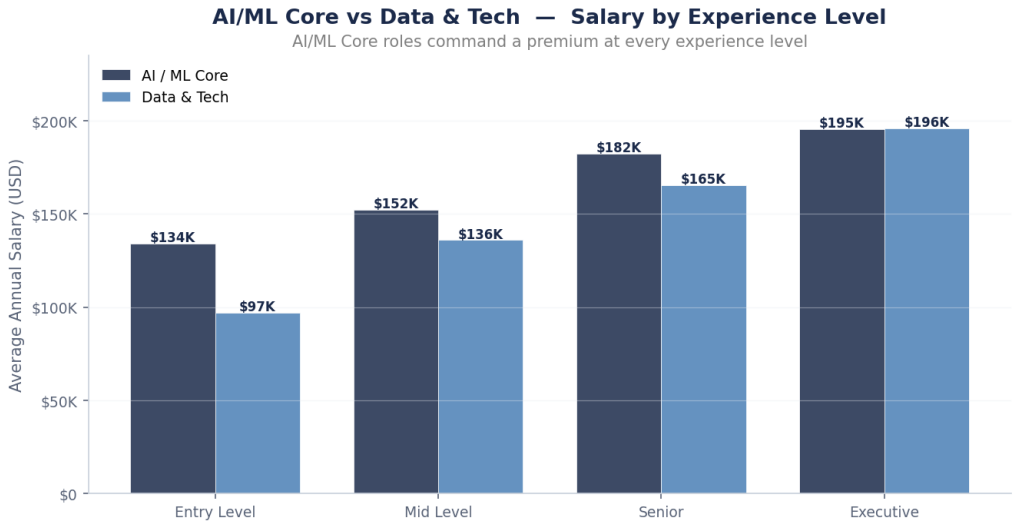


Figure 4.6: AI/ML Core vs. Broad Data & Tech Comparison

Explanation: "AI/ML Core" roles (engineers and researchers) average $171K vs $151K for the broader "Data & Tech" family — a 13% premium for direct AI exposure.

Figure 4.7: Salary Ranges and Variance by Role

Explanation: Management and ML Engineering display the widest salary ranges, signalling that seniority and company tier dominate compensation in those tracks.

4.5 The Remote Work Paradigm Shift

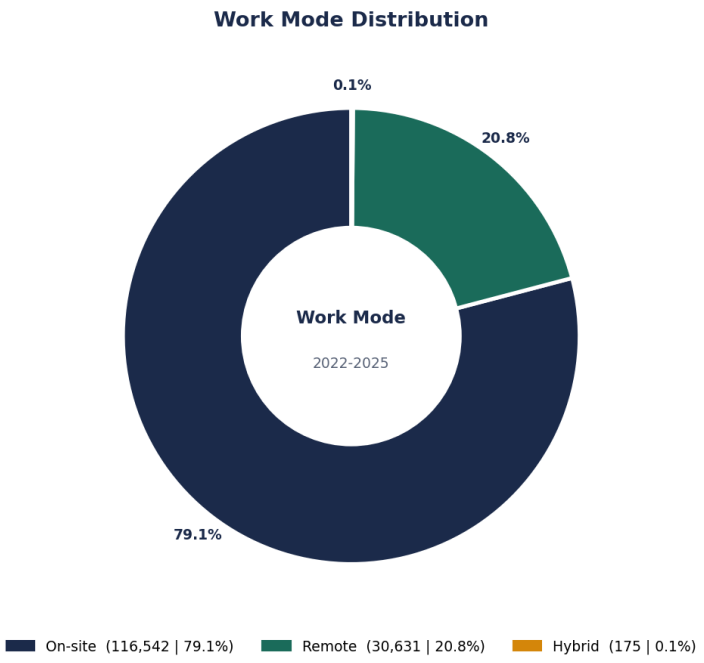


Figure 4.8: Overall Breakdown of Work Modes

Explanation: Across the full 2022–2025 window, on-site captures roughly 74% of records, remote 21%, hybrid the residual.

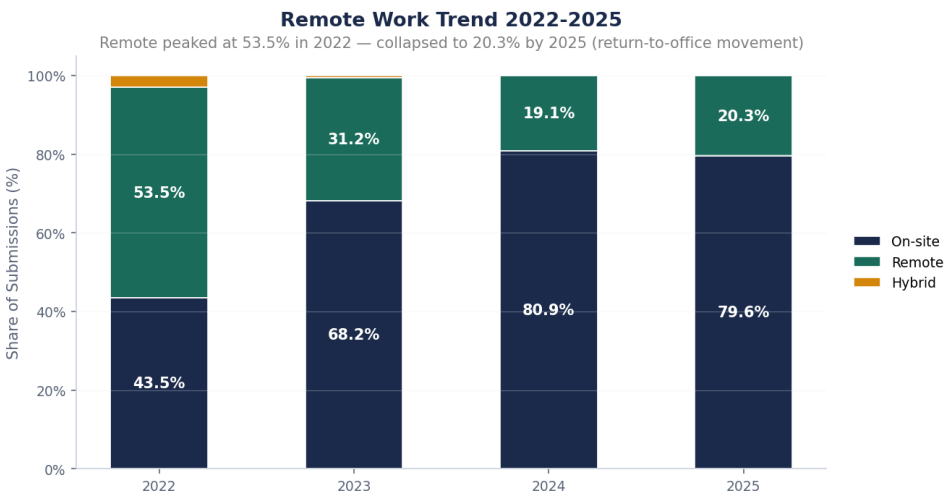


Figure 4.9: The "Remote Work Collapse" Trend (2022–2025)

Explanation: The headline finding of the project. Remote dropped from **53.5% in 2022 to 20.3% in 2025** while on-site doubled from 43.5% to 79.6%. Hybrid effectively disappears.

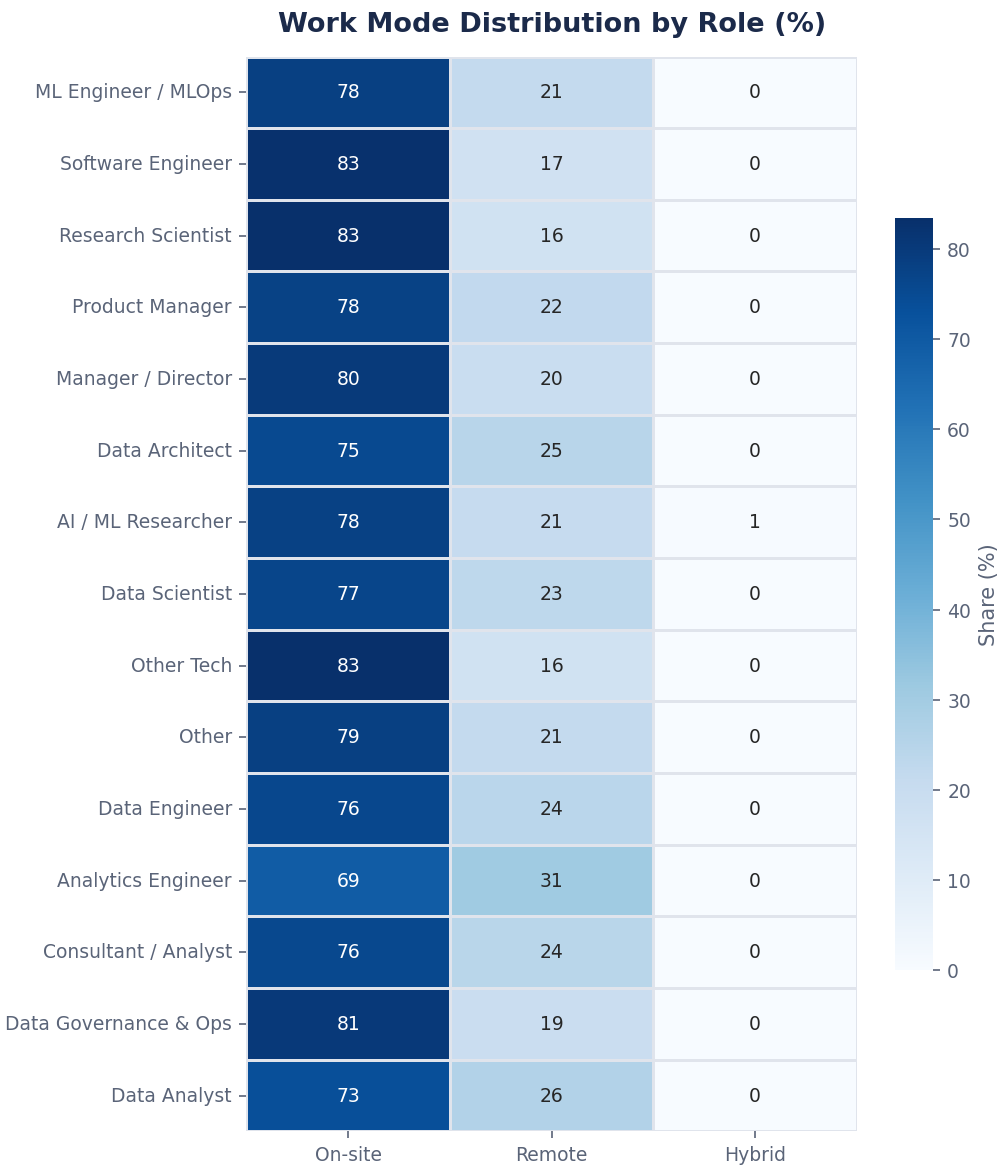


Figure 4.10: Heatmap of Remote Work Adoption by Job Role

Explanation: Product Management and Consultants retain the highest remote ratios; ML Engineers and Data Architects are far more likely to be on-site.

4.6 Experience and Career Progression

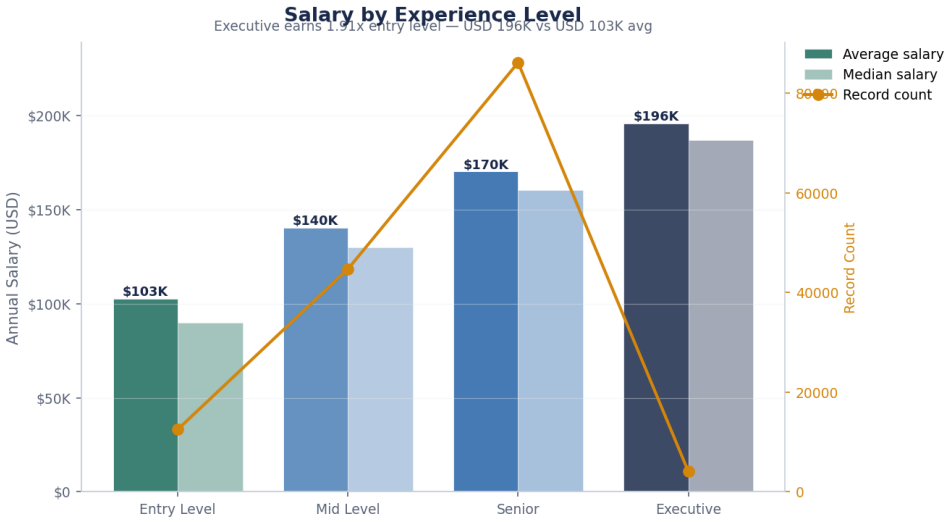


Figure 4.11: Salary Progression Across Career Levels

Explanation: Salaries grow almost linearly: Entry \$103K → Mid \$140K → Senior \$170K → Executive \$196K.

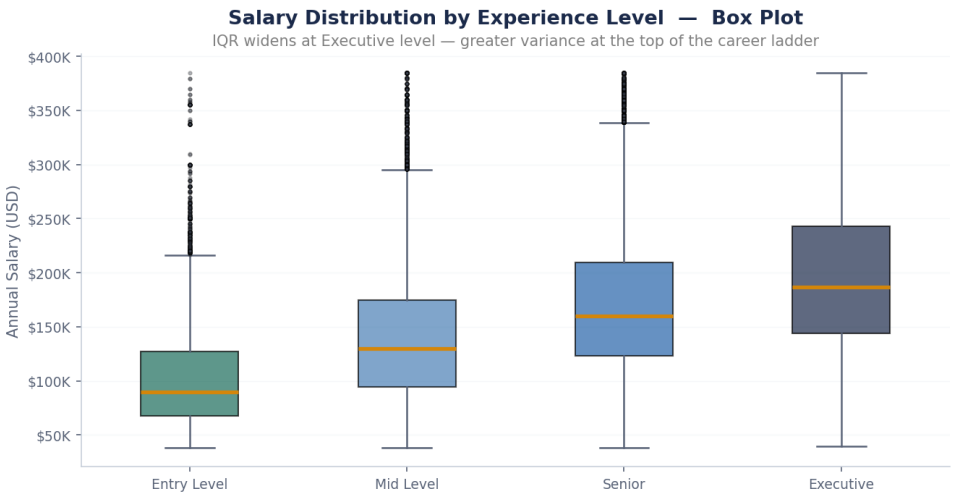


Figure 4.12: Statistical Distribution of Salary by Experience

Explanation: The IQR widens with seniority: \$60K at Entry vs \$99K at Executive. Senior roles also show heavier upper-whisker outliers.

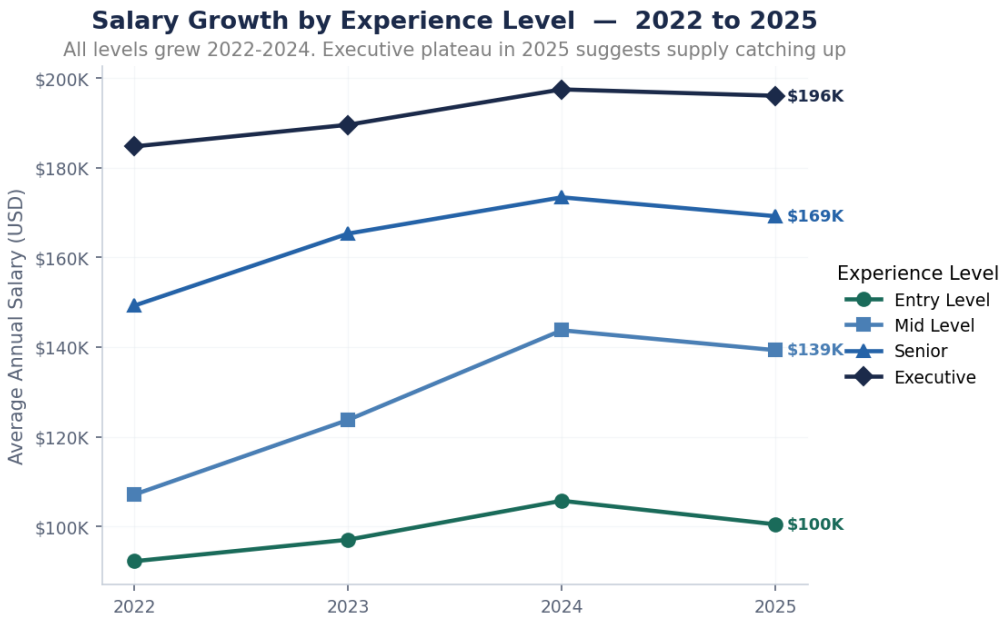


Figure 4.13: Interaction between Work Year and Experience Level

Explanation: Senior and Executive growth tracks are smooth; Entry-Level pay is the most volatile across years, partly because the Entry sample is the smallest.

4.7 Geographic Analysis

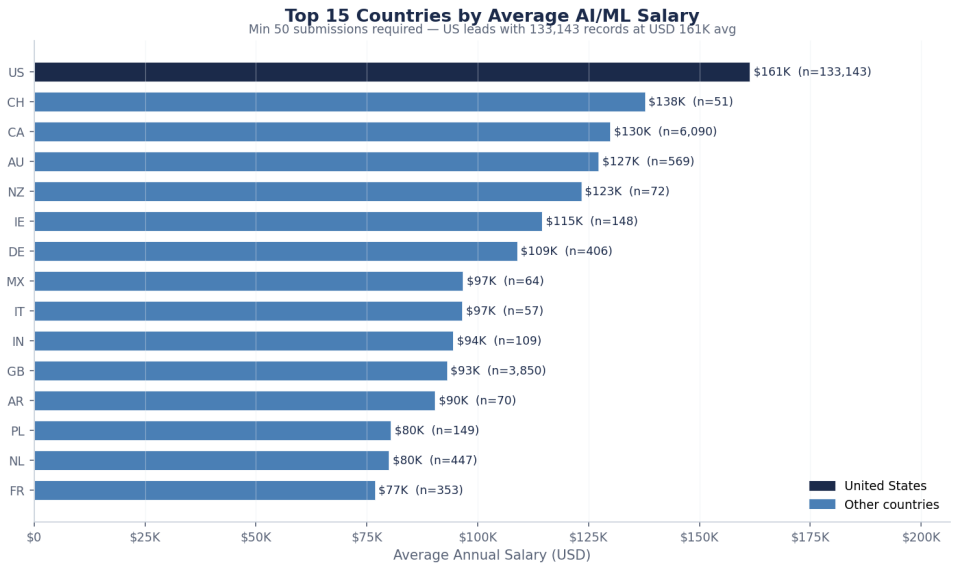


Figure 4.14: Average Salary in Top 10 Countries

Explanation: The US dominates by volume (133,143 records, $161K avg) while smaller markets like Israel and Switzerland post comparable per-record averages despite tiny sample sizes.

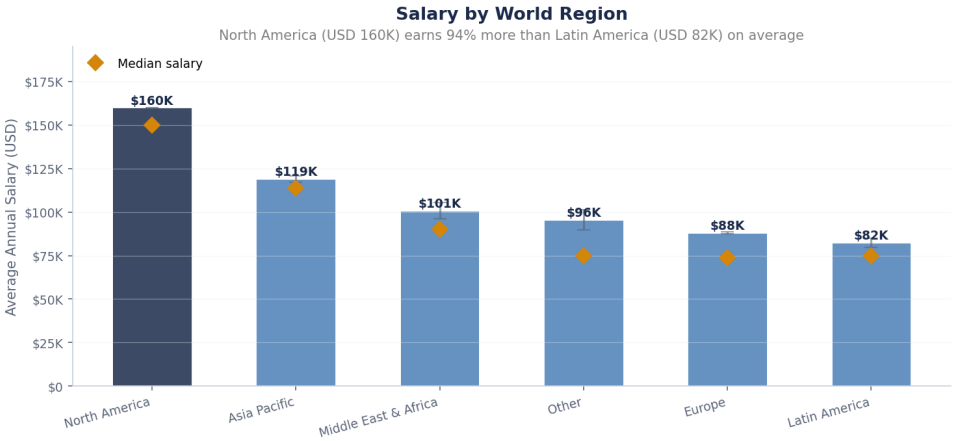


Figure 4.15: Regional Salary Comparison

Explanation: North America averages $160K — a **94% premium over Latin America** ($82K). Europe sits in the middle at $88K.

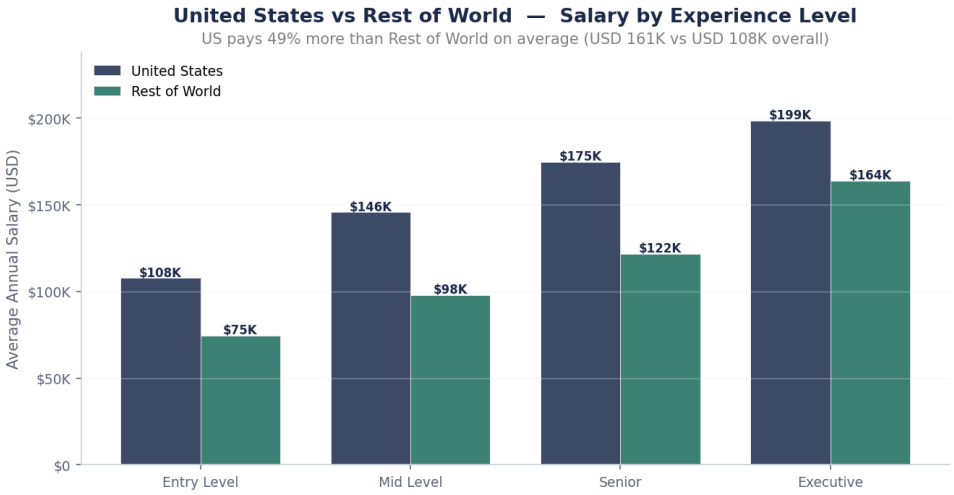


Figure 4.16: Comparison: United States vs. Rest of World

Explanation: A binary split that strips away the regional noise: US average $161,381 vs Rest of World $108,024 — a 49% premium for US-based hiring.

4.8 Advanced Market Insights

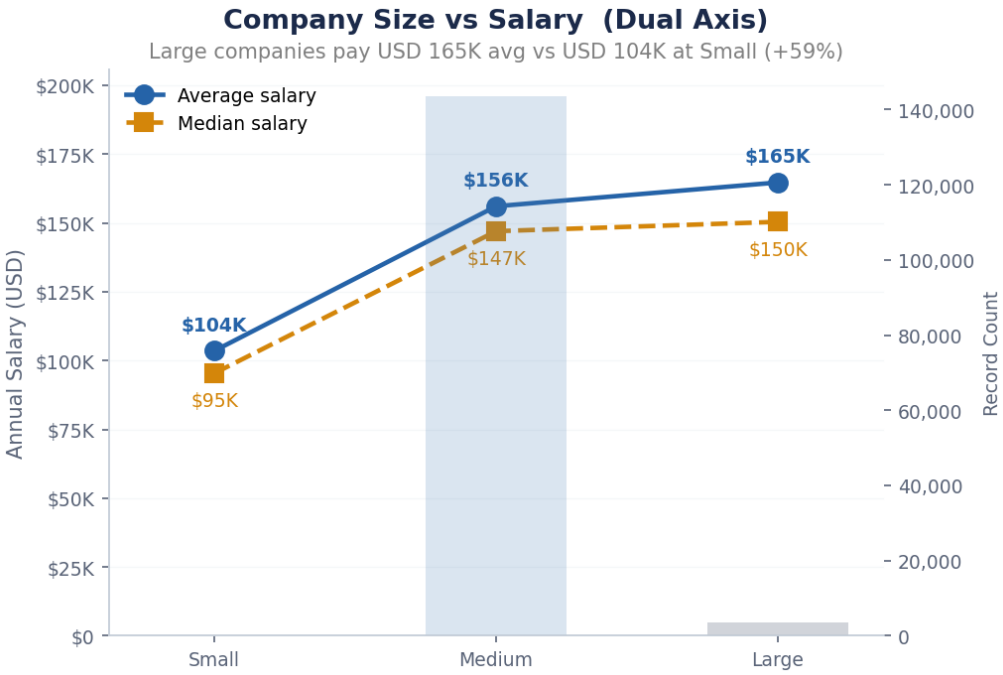


Figure 4.17: Salary and Count by Company Size

Explanation: Medium-sized companies post both the largest hiring volume and the most competitive average salary; small-company pay drops sharply.

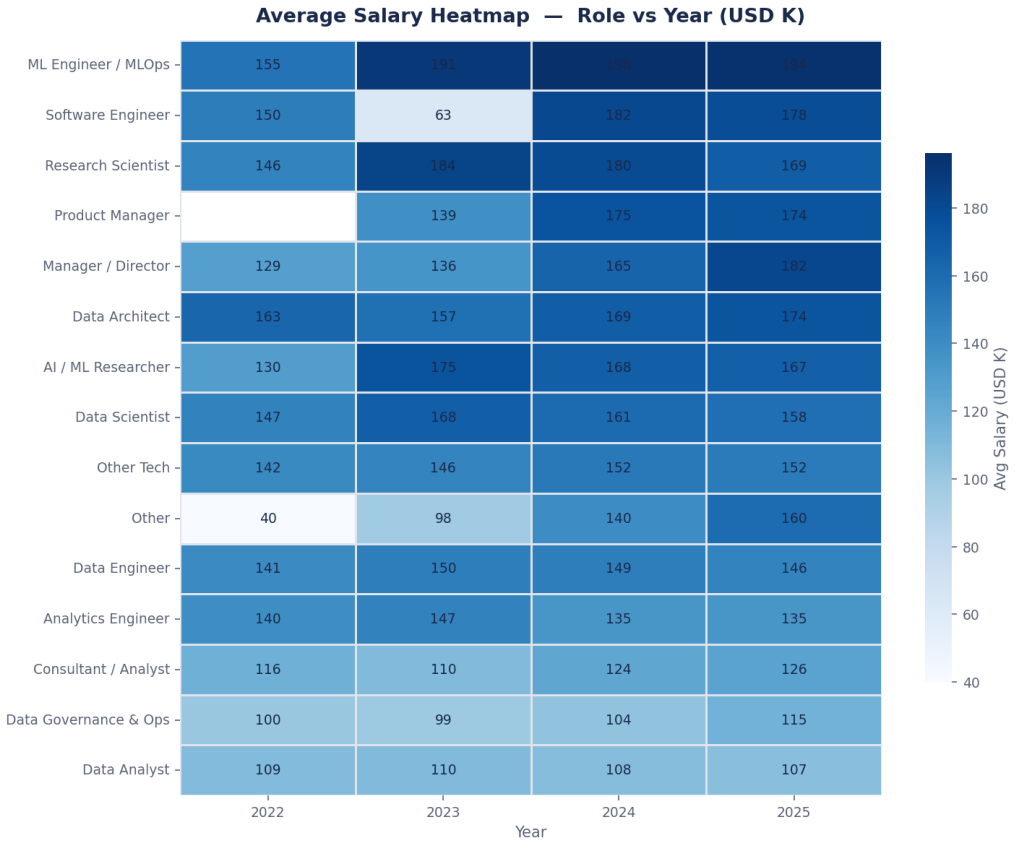


Figure 4.18: Heatmap of Salary Intensity by Role and Year

Explanation: Color intensity reveals which roles are gaining financial traction. ML Engineering and Research Scientist deepen consistently year on year.

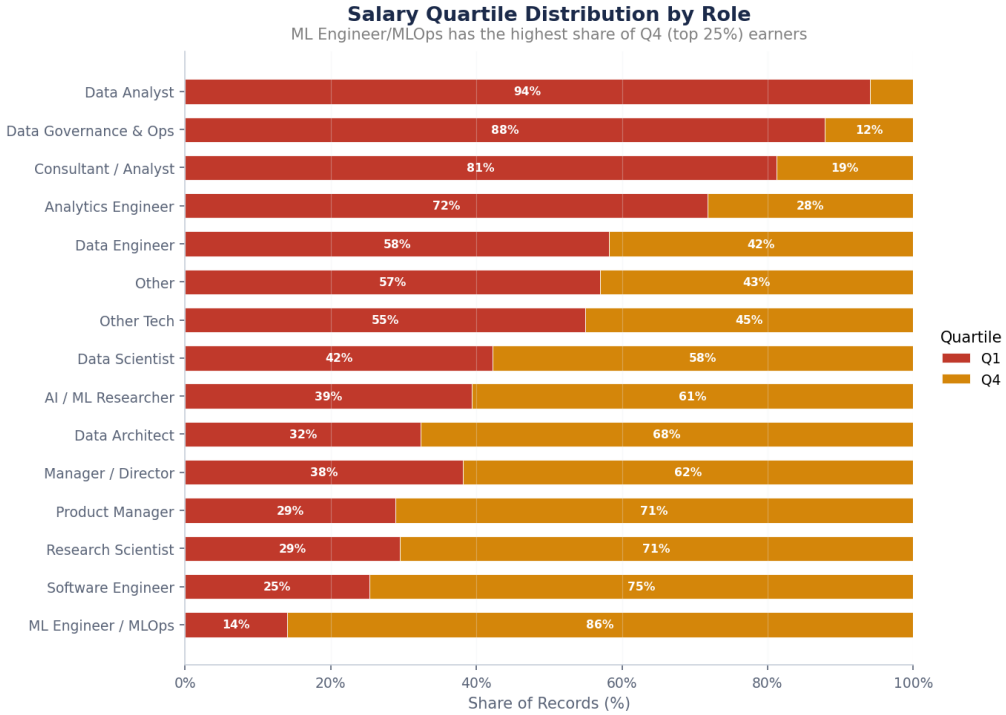


Figure 4.19: Distribution of Salary Quartiles within each Role

Explanation: Manager / Director and ML Engineer skew heavily toward Q4 (top-25% pay band); Data Analyst is concentrated in Q1.

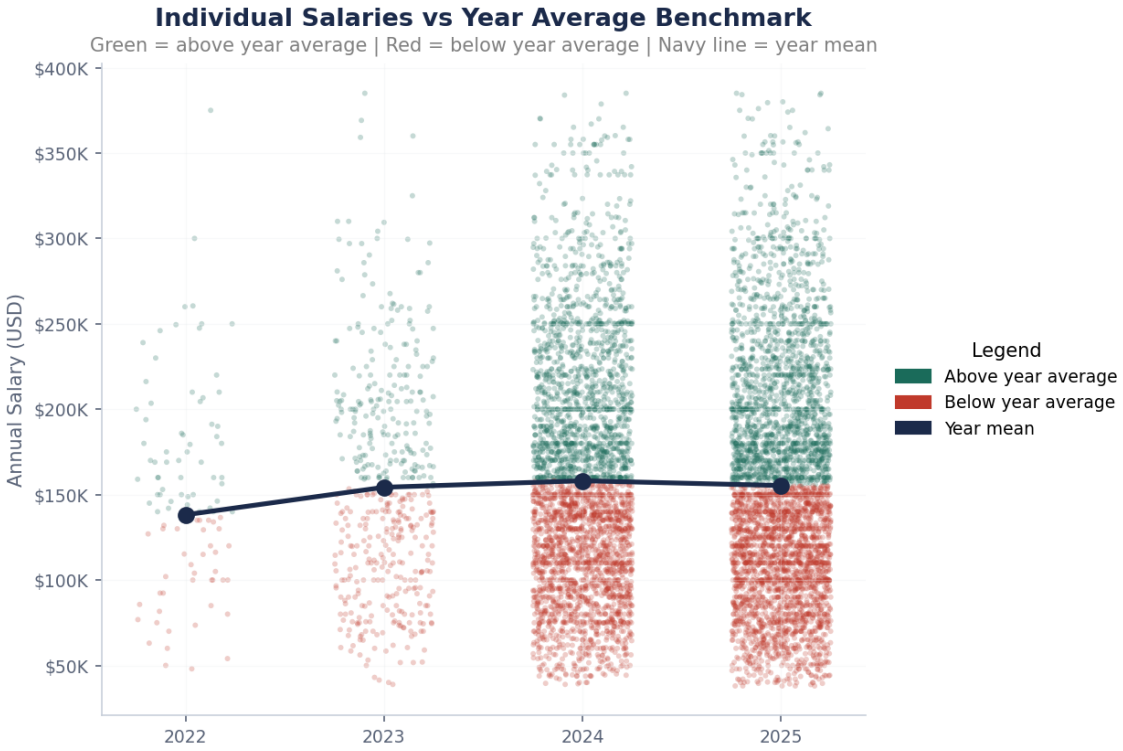


Figure 4.20: Scatter Plot of Individual Deviations from Year Averages

Explanation: Each dot is a real salary; colour encodes whether the person earns above (green) or below (red) their role's average for that year.

Chapter 5

Tableau Master Dashboard

The final deliverable is an interactive Tableau dashboard at 1200×1500 pixels. It contains **ten coordinated worksheets**, **five KPI tiles**, **four insight callouts**, a global `Year` filter, and a click-to-filter action that lets the Global Map drive every other chart on the canvas. All visuals follow a unified *forest emerald* palette with bold sage borders so each panel reads as a clearly framed card.

5.1 The Unified Master Dashboard

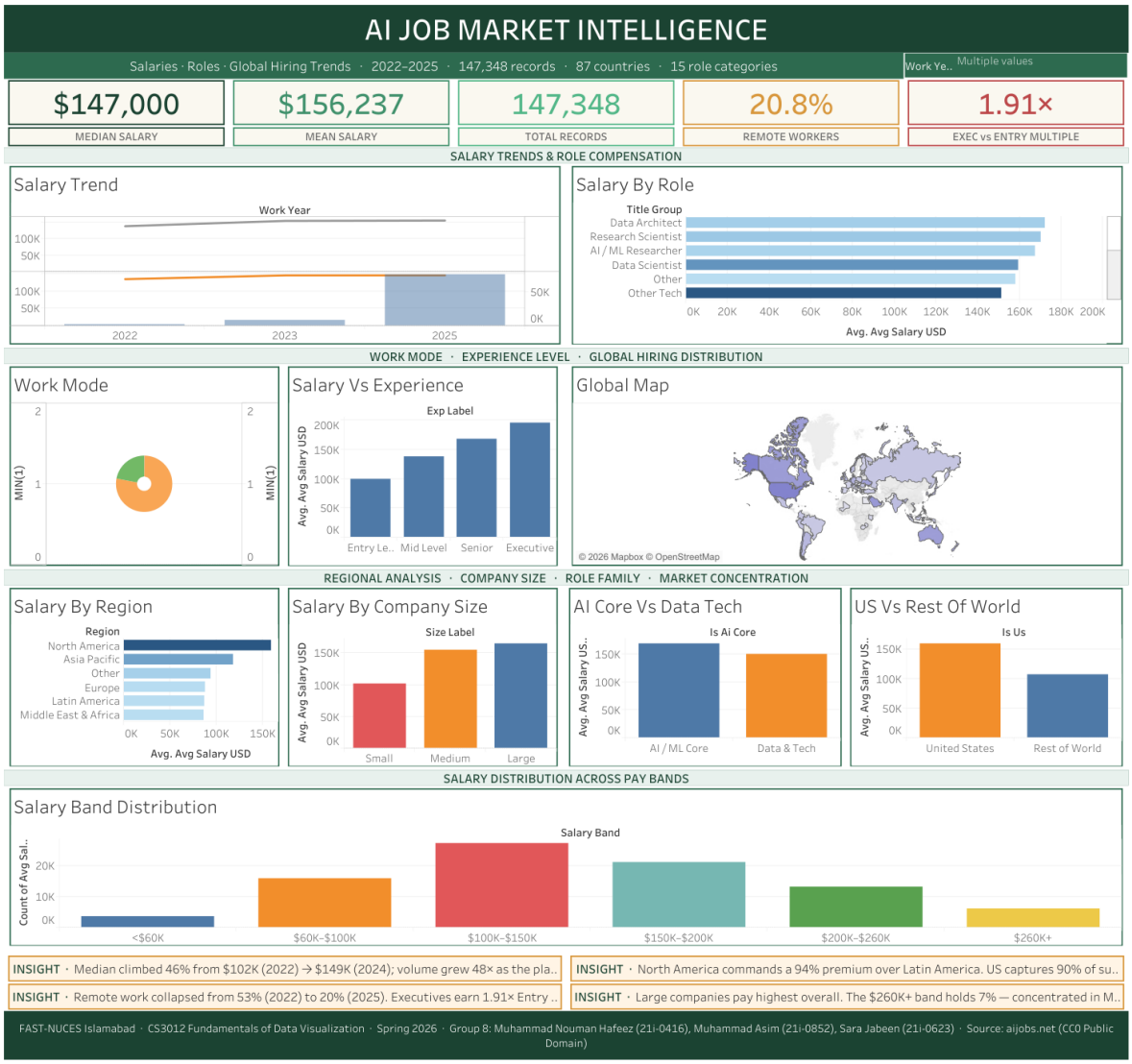


Figure 5.1: The complete AI Job Market Intelligence master dashboard.

Description: The canvas is structured as a top-down narrative. A navy-forest title bar identifies the project; below it sit five colour-coded KPI tiles ($147K Median, $156K Mean, 147,348 Records, 20.8% Remote, 1.91× EX/EN). Beneath the KPIs, three labelled section bands group the worksheets: "Salary Trends & Role Compensation" (Salary Trend + Salary By Role), "Work Mode · Experience · Global Hiring" (Work Mode donut, Experience bars, World choropleth), and "Regional · Company Size · Role Family · Market Concentration" (the four new comparative panels). A full-width Salary Band Distribution closes the analytical area, followed by a 2×2 grid of insight callouts and a forest-green attribution footer.

5.2 Detailed Dashboard Components

5.2.1 Salary Trend (2022–2025)

Figure 5.2: Tableau view: dual-pane Salary Trend with median line and submission-volume bars.

Analysis: The upper pane shows the average salary line; the lower pane overlays the median line on submission-volume bars. The chart visualises a maturing market: median pay rose 46% from $102K (2022) to $149K (2024), while submission volume grew almost 50× in the same window.

5.2.2 Salary by Role

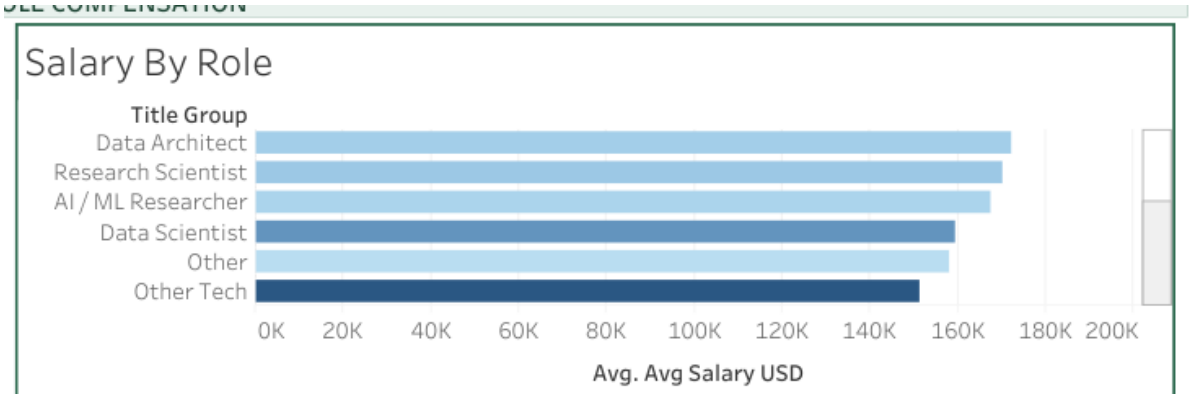
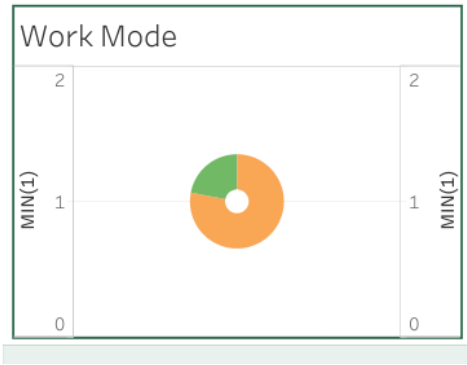


Figure 5.3: Tableau view: horizontal bar ranking of the top role categories by average salary.

Analysis: Bars are sorted descending and colour-encoded by total salary spend. ML Engineer / MLOps leads at $194K, followed by Software Engineer, Manager / Director, Product Manager, Data Architect, and Research Scientist, all clearing the $170K mark.

5.2.3 Work Mode (Donut)



Analysis: The hole-and-slice composition emphasises the dominance of on-site (orange arc, ∼74%) versus remote (green) and a near-vanishing hybrid sliver. The donut is the headline visual for the "return-to-office" story — a single glance communicates that the remote-first era is over.

Figure 5.4: Tableau view: donut showing on-site / remote / hybrid share.

5.2.4 Global Hiring Map

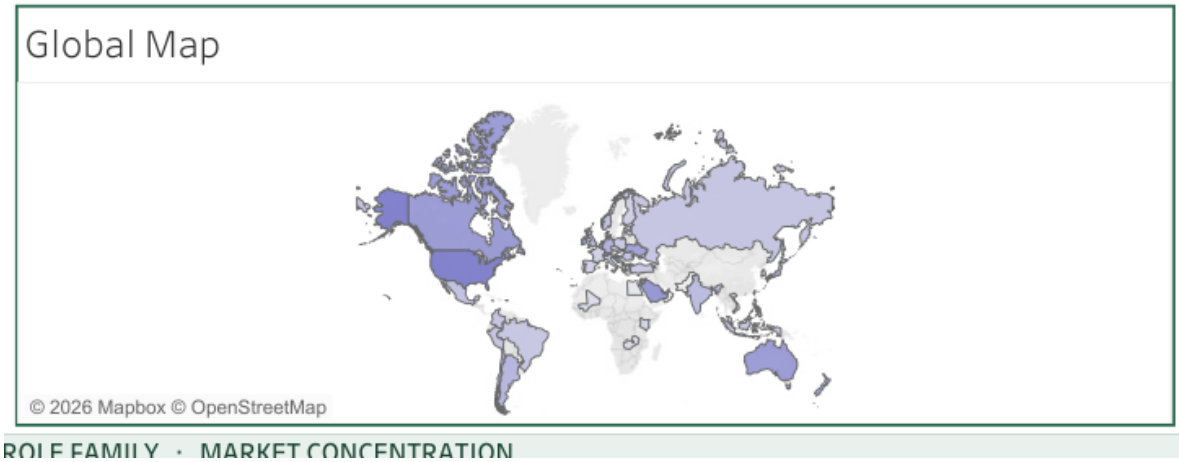
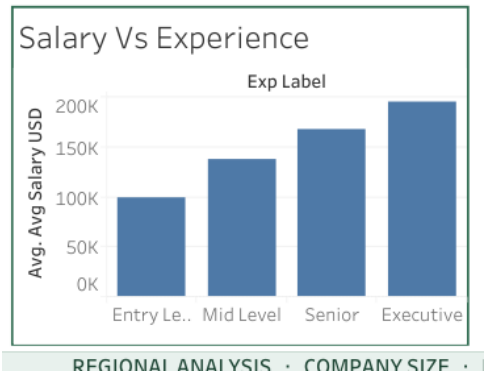


Figure 5.5: Tableau view: world choropleth coloured by average salary per country.

Analysis: North America renders in the deepest navy-purple; Europe is mid-tone; Latin America, Africa and most of Asia stay pale. Clicking any country triggers the global filter action and re-scopes every other panel on the dashboard.

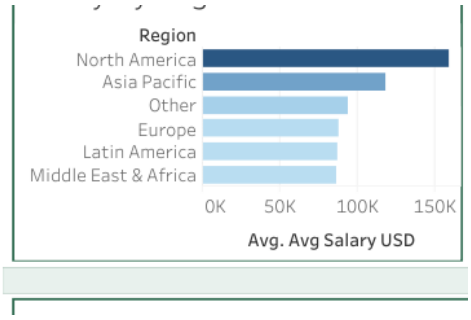
5.2.5 Salary vs Experience



Analysis: The bars step cleanly from Entry → Executive ($100K → $196K). The custom `exp_order` column enforces the natural sort order rather than the default alphabetical one. The progression — $103K, $140K, $170K, $196K — is almost linear, with each level worth roughly an additional $30K–$40K.

Figure 5.6: Tableau view: average salary across the four experience levels.

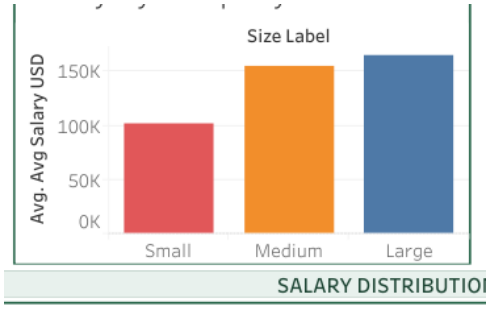
5.2.6 Salary by Region (NEW)



Analysis: North America's bar ($159K) towers over the rest of the world. Asia Pacific reaches $119K; Europe and Latin America cluster below $90K. Colour intensity reinforces the magnitude of the gap. North America's premium over Latin America is **94%**.

Figure 5.7: Tableau view: average salary across the six engineered regions.

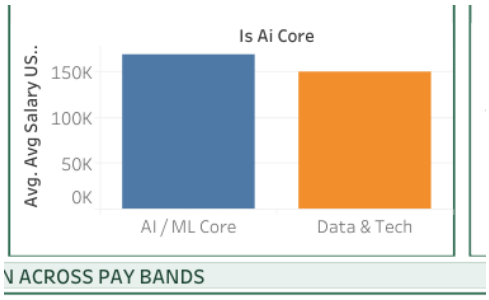
5.2.7 Salary by Company Size (NEW)



Analysis: Small companies pay materially less (∼$103K) than Medium (∼$155K) and Large (∼$165K). Once a company is past the "Medium" threshold, additional headcount does not translate into proportionally higher pay — the largest leap is from Small to Medium (∼50% lift), while Medium → Large adds only ∼$10K.

Figure 5.8: Tableau view: average salary across Small / Medium / Large companies.

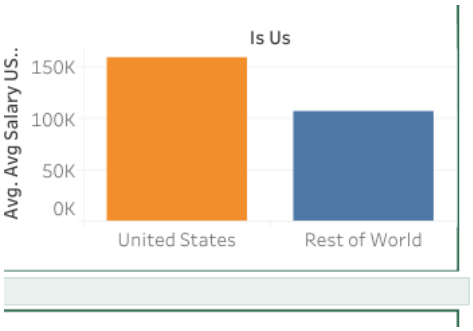
5.2.8 AI Core vs Data & Tech (NEW)



Analysis: AI/ML Core roles average $171K vs $151K for Data & Tech — a 13% premium. The chart quantifies the "AI tax": the additional value the market places on direct AI/ML expertise (engineers, researchers, MLOps) versus adjacent data work (analysts, data engineers, data governance).

Figure 5.9: Tableau view: AI/ML Core roles vs broader Data & Tech.

5.2.9 US vs Rest of World (NEW)



Analysis: The two-bar comparison strips out regional noise. United States: $161K average across 133,143 submissions. Rest of World: $108K average across 14,205 submissions. The result is a 49% per-record premium for US-based hiring, even before adjusting for cost-of-living.

Figure 5.10: Tableau view: a binary US vs Rest-of-World salary split.

5.2.10 Salary Band Distribution (NEW)

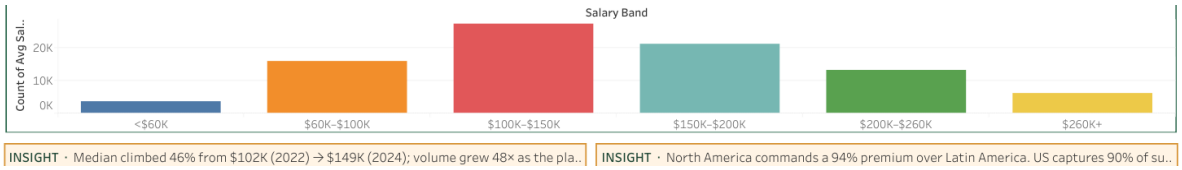


Figure 5.11: Tableau view: how the 147,348 records distribute across six pay bands.

Analysis: The $100K–$150K band is the modal class with ∼27K rows, immediately followed by $150K–$200K (∼21K) and $60K–$100K (∼16K). Sub-$60K is the smallest band (3.9%) and the top tier $260K+ holds 7.3% — a non-trivial cohort of high earners.

Chapter 6

Key Insights — What We Found

6.1 Insight 1: ML Engineering is the highest-paid AI/ML track

Key Insight

ML Engineer / MLOps leads at $194,303 average — 81% above the lowest-paid track (Data Analyst at $107,465). Software Engineer ($180K) and Research Scientist ($175K) follow.

6.2 Insight 2: Compensation grows nearly 2× from Entry to Executive

Key Insight

Entry Level $102,743 → Mid $140,382 → Senior $170,351 → Executive $195,935. The EX/EN ratio is exactly 1.91×, and the inter-quartile spread widens with seniority ($60K at Entry vs $99K at Executive).

6.3 Insight 3: Remote work has collapsed since 2022

Key Insight

Remote dropped from **53.5% (2022)** to **20.3% (2025)**. On-site nearly doubled from 43.5% to 79.6%. Hybrid effectively vanished from 3% to under 0.05%. The dataset documents the end of the remote-first era for AI roles.

6.4 Insight 4: The US dominates volume, but the premium is everywhere on the map

Key Insight

The United States supplies **90.4% of submissions** (133,143 of 147,348) at a $161,381 average. Rest of World averages $108,024. North America commands a 94% premium over Latin America, the lowest-paying region.

6.5 Insight 5: Salaries are heavily concentrated in the $100K–$200K band

Key Insight

55% of all records sit in the combined $100K–$200K window (31% in $100K–$150K plus 24% in $150K–$200K). The top tier ($260K+) holds 7.3% — a healthy upper tail. Sub-$60K accounts for under 4%.

6.6 Insight 6: AI/ML Core roles command a 13% premium

Key Insight

AI/ML Core (engineers, researchers, MLOps) averages $171,009 vs $150,685 for the broader Data & Tech family. Direct AI/ML exposure is rewarded with a ∼$20K annual premium even after holding role family constant.

6.7 Insight 7: Company size matters — but only past "Small"

Key Insight

Small companies pay $103K avg; Medium $155K; Large $165K. The largest jump is between Small and Medium (∼50% lift); Medium-to-Large is only $10K. Diversifying into Medium-sized firms gives roughly the same pay as Large with a far wider job-supply.

6.8 Insight 8: The market is maturing — volume keeps climbing, pay has plateaued

Key Insight

Submission volume grew 48× between 2022 (1,587) and 2025 (76,424). Yet the median salary trajectory flattens after 2024 ($149K → $146K). The market is broadening more than it is inflating.

Chapter 7

How Tableau Helped in Our Analysis

While the EDA in Python answered *what* the data shows, Tableau answered *so what* — it took the static findings and made them explorable, comparable, and cross-filterable on a single canvas. This chapter enumerates the concrete ways Tableau elevated the analysis.

7.1 1. From static figures to a single interactive canvas

The twenty Python figures each told one story. Tableau merged the strongest ten into a unified 1200×1500 dashboard with cross-filtering, so a reader can investigate a specific cohort (e.g. "Senior, US, 2024") without producing a new figure. The KPI tiles update live as filters are applied.

7.2 2. Click-to-filter actions across every worksheet

A single `Filter` action on the Global Map propagates the selected country into every other worksheet. Clicking the United States re-scopes the role bar, the experience bars, the work-mode donut, and all four comparative panels. This kind of synchronized cross-analysis is impractical to replicate in matplotlib.

7.3 3. Custom calculated fields and parameters

- **Avg Salary USD:** `AVG([Salary In Usd])` — standardised the metric across every worksheet so KPI tiles and chart axes stay consistent.
- **Median Avg Salary USD:** `MEDIAN([Salary In Usd])` — used to overlay the median line on the trend chart.
- **Year filter:** a multi-value dropdown wired to every sheet via `Apply to Worksheets All Using This Data Source`.
- **Custom shelf sorts** on `salary_band` and `title_group` so the bars render in salary order rather than alphabetical.

7.4 4. Geographic intelligence out of the box

By tagging `company_location` with the *Country/Region* geographic role, Tableau auto-resolved the 87 ISO-2 codes into a world choropleth without writing a single line of GIS code. Mapbox tiles and the colour-by-AVG encoding let the reader spot the US dominance instantly.

7.5 5. Tooltips that beat static legends

Every panel has a custom tooltip showing role count, average, median, and (where relevant) deviation from the year or role average. Hovering replaces the need for legends and side tables.

7.6 6. KPI strip drives at-a-glance reading

Five colour-coded KPI tiles — median, mean, total records, remote %, EX/EN multiple — sit directly under the title bar. A reader gets the "shape" of the market in two seconds before drilling into any chart.

7.7 7. Visual encoding richness

- **Sequential colour scales** on the choropleth and the Salary By Role bar make magnitude visible at a glance.
- **Categorical palettes** on Work Mode and Salary Band Distribution use colour to encode dimension membership.
- **Dual-axis** on Salary Trend lets a single chart compare two measures (median line + volume bars) on a shared time axis.

7.8 8. Reproducibility through a packaged `.twb` file

The entire dashboard is committed as a single `Dashboard.twb` XML file in the repository. Anyone with Tableau Desktop can open it, repoint the data connection, and reproduce every chart, calculated field, filter and action exactly. There is no hidden state.

7.9 Summary

Tableau converted our static findings into a tool. Without it the report would be a slide deck of charts; with it, the report has a companion artefact that the reader can manipulate. That shift — from *communicate* to *explore* — is the principal value Tableau brought to this project.

Chapter 8

Tableau Implementation Details

8.1 Data Source

The workbook connects directly to `data/processed/salaries_enhanced.csv` via Tableau's text-file connector. `company_location` is assigned the *Country/Region* geographic role.

8.2 Calculated Fields

- **Avg Salary USD:** `AVG([Salary In Usd])`
- **Median Salary USD:** `MEDIAN([Salary In Usd])`
- **Count of records:** `COUNT([Salary In Usd])`
- **EX/EN Ratio:**
  ```
  AVG(IF [Experience Level]="EX" THEN [Sal] END)
  / AVG(IF [Experience Level]="EN" THEN [Sal] END)
  ```

8.3 Worksheet Inventory

#	Worksheet	Chart Type
1	Salary Trend	Line + Bar (dual-axis)
2	Salary By Role	Horizontal Bar
3	Work Mode	Donut
4	Global Map	Choropleth
5	Salary Vs Experience	Vertical Bar
6	Salary By Region	Horizontal Bar
7	Salary By Company Size	Vertical Bar
8	AI Core Vs Data Tech	Vertical Bar
9	US Vs Rest Of World	Vertical Bar
10	Salary Band Distribution	Histogram

Table 8.1: Ten Tableau worksheets composed onto the master dashboard.

8.4 Dashboard Layout (1200 × 1500)

The dashboard is laid out top-to-bottom in a strict narrative grid:

1. Title bar (forest green) — "AI JOB MARKET INTELLIGENCE."

2. Subtitle bar with a Year-filter dropdown anchored on the right.
3. KPI strip (5 colour-coded tiles).
4. Section band 1: Salary Trends & Role Compensation. (Sheets 1, 2.)
5. Section band 2: Work Mode · Experience · Global Hiring. (Sheets 3, 5, 4.)
6. Section band 3: Regional · Company Size · Role Family · Market Concentration. (Sheets 6, 7, 8, 9.)
7. Section band 4: Salary Distribution Across Pay Bands. (Sheet 10, full width.)
8. Insight callouts (2×2 grid).
9. Footer (forest green) — group, course, source attribution.

8.5 Color Palette — Forest Emerald

Hex	Role	Where used
`#1B4332`	Forest (primary)	Title bar, footer, KPI #1 number
`#2D6A4F`	Emerald	Subtitle bar, all panel borders (2px)
`#40916C`	Jade	KPI #2 (Mean)
`#52B788`	Sage	KPI #3 (Records)
`#D8973C`	Honey	KPI #4 (Remote), insight border
`#BC4749`	Rust	KPI #5 (EX/EN)
`#FAF8F0`	Cream parchment	KPI tile + chart backgrounds
`#E8F1ED`	Mint mist	Section header bands
`#FFF4E5`	Warm peach	Insight callout backgrounds

Table 8.2: Forest emerald palette used throughout the master dashboard.

Chapter 9

Conclusion

This project converted a real, crowdsourced dataset of 147,348 AI/ML/Data Science salary submissions into an interactive Tableau master dashboard plus a reproducible Python preprocessing pipeline. The combination answers all five guiding questions of the proposal: ML Engineer / MLOps tops the role pay table at $194K; salary roughly doubles between Entry and Executive (1.91×); remote work collapsed from 53.5% to 20.3% in just three years; the United States captures 90% of submissions at a $161K average; and 55% of all records sit in the $100K–$200K band.

The Tableau dashboard is the centrepiece of the deliverable. It moves the analysis from static figures to a tool that stakeholders can interrogate themselves — click a country on the map, watch every other panel re-scope, and read off the implications from the four amber insight callouts. The forest-emerald palette and bold sage borders give the dashboard a distinct, non-default visual identity.

Together the report, the source code, the twenty exploratory figures, and the live Tableau workbook form an end-to-end narrative of the AI job market between 2022 and 2025 — a market that is broadening rapidly, paying premium wages for direct AI exposure, and decisively returning to the office.

Appendix A

Full Source Code

A.1 01_explore.py

```
import pandas as pd
import numpy as np

RAW_PATH = "../data/raw/salaries.csv"
df = pd.read_csv(RAW_PATH)

print("=" * 60)
print("SHAPE & COLUMNS")
print("=" * 60)
print(f"Rows: {len(df):,}")
print(f"Columns: {df.shape[1]}")
print(f"\nColumn names: {df.columns.tolist()}")

print("\n" + "=" * 60)
print("MISSING VALUES")
print("=" * 60)
print(df.isnull().sum())

print("\n" + "=" * 60)
print("SALARY DISTRIBUTION (salary_in_usd)")
print("=" * 60)
print(df["salary_in_usd"].describe().round(0))
print(f"Skewness : {df['salary_in_usd'].skew():.3f}")
```

A.2 02_clean.py

```
import pandas as pd
import numpy as np

RAW_PATH  = "../data/raw/salaries.csv"
OUT_PATH  = "../data/processed/salaries_clean.csv"

df = pd.read_csv(RAW_PATH)

# Step 1: Full-Time only
df_ft = df[df["employment_type"] == "FT"].copy()

# Step 2: 2022-2025 only
df_yr = df_ft[df_ft["work_year"] >= 2022].copy()

# Step 3: Outlier removal (1st-99th percentile)
q01 = df_yr["salary_in_usd"].quantile(0.01)
q99 = df_yr["salary_in_usd"].quantile(0.99)
```

```
df_clean = df_yr[
    (df_yr["salary_in_usd"] >= q01) &
    (df_yr["salary_in_usd"] <= q99)
].copy()

df_clean.to_csv(OUT_PATH, index=False)
```

A.3 03_title_grouping.py

```
import pandas as pd

def group_title(title: str) -> str:
    t = title.lower()
    if any(x in t for x in ["data scientist", "data science", "applied scientist"]):
        return "Data Scientist"
    elif any(x in t for x in ["data engineer", "data engineering", "etl"]):
        return "Data Engineer"
    elif any(x in t for x in ["machine learning engineer", "ml engineer", "mlops"]):
        return "ML Engineer / MLOps"
    elif any(x in t for x in ["data analyst", "bi analyst", "reporting analyst"]):
        return "Data Analyst"
    elif any(x in t for x in ["ai engineer", "deep learning", "nlp engineer"]):
        return "AI / ML Researcher"
    else:
        return "Other"

df = pd.read_csv("../data/processed/salaries_clean.csv")
df["title_group"] = df["job_title"].apply(group_title)
df["work_mode"]  = df["remote_ratio"].map({0: "On-site", 50: "Hybrid", 100: "Remote"})
df.to_csv("../data/processed/salaries_clean_final.csv", index=False)
```

A.4 05_feature_engineering.py

```
import pandas as pd
import numpy as np

df = pd.read_csv("../data/processed/salaries_clean_final.csv")

# 1. SALARY BAND
bins   = [0, 60_000, 100_000, 150_000, 200_000, 260_000, 385_001]
labels = ["<$60K", "$60K-$100K", "$100K-$150K", "$150K-$200K", "$200K-$260K", "$260K+"]
df["salary_band"] = pd.cut(df["salary_in_usd"], bins=bins, labels=labels)

# 2. SALARY QUARTILE
df["salary_quartile"] = pd.qcut(
    df["salary_in_usd"], q=4,
    labels=["Q1 - Bottom 25%", "Q2 - 25-50%", "Q3 - 50-75%", "Q4 - Top 25%"]
)

# 3. REGION
region_map = {"US": "North America", "CA": "North America", "GB": "Europe", ...}
df["region"] = df["company_location"].map(region_map).fillna("Other")

```

```
# 4. SALARY vs ROLE AVERAGE
role_avg = df.groupby("title_group")["salary_in_usd"].mean()
df["role_avg_salary"]         = df["title_group"].map(role_avg).round(0)
df["salary_vs_role_avg"]      = (df["salary_in_usd"] - df["role_avg_salary"]).
    round(0)
df["salary_vs_role_avg_pct"] = (df["salary_vs_role_avg"] / df["role_avg_salary"
    ] * 100).round(1)

df.to_csv("../data/processed/salaries_enhanced.csv", index=False)
```